Math Kangaroo Practice - Grades 3-4

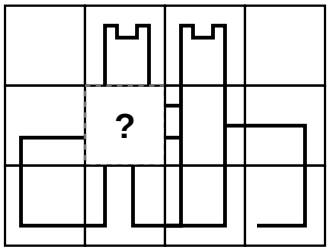
Practice set for 2002 #1

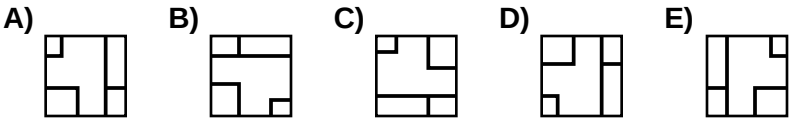
Five practice problems per selected problem.

Set 1 - Missing puzzle tile

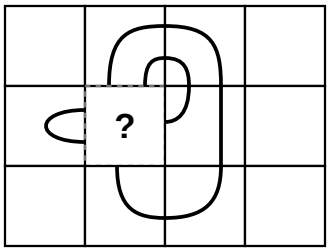
Based on 2002, Problem 1

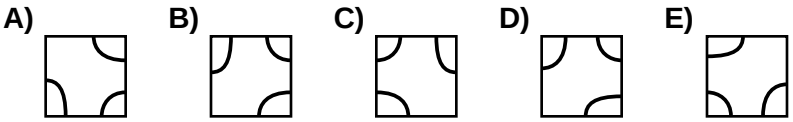
Practice 1.1 Which square tile (A through E) is missing from the castle drawing below?



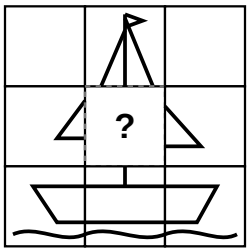


Practice 1.2 A model train track loops across the 3 × 4 board below. Which tile (A through E) belongs in the missing square marked with a question mark (?) to make the track continuous?



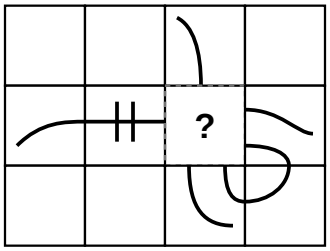


Practice 1.3 Which of the squares below is missing from the picture of the sailboat?



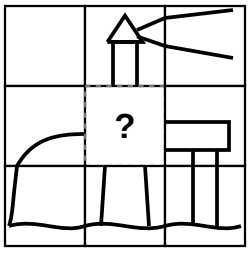
A)
B)
C)
D)
E)

Practice 1.4 Streams flow across the 3 × 4 map below, and one square tile is missing. Which tile (A through E) fits in the square marked with a question mark (?)?



A)
B)
C)
D)
E)

Practice 1.5 Which of the squares below is missing from the picture of the lighthouse?



A)
B)
C)
D)
E)

Answer Key & Explanations

For generated/2026-10-19-20-26-07.pdf (Practice set for 2002 #1)

Solutions & kid-friendly explanations

1.1: E In the missing square, the left turret wall runs straight from 0.25 at the top to 0.25 at the bottom (with the outer wall meeting it from the left at 0.35), while the right turret wall enters at 0.80 from the top to meet the upper bridge at 0.75 on the right, and the lower gate rises from 0.60 at the bottom to 0.35 on the right. Only tile E fits.

1.2: C Tracing the tracks into the missing square: two tracks enter from the left (at 0.30 and 0.70), two enter from the top (at 0.30 and 0.75), one exits at the bottom (0.40), and one exits at the right (0.55). Only tile C connects every track without breaks.

1.3: C In the center square, the vertical mast runs straight from the top center (0.50) to the bottom center (0.50), the left jib sail is higher than the right mainsail, and both horizontal booms meet the mast. Only square C matches all four edges of the missing center square.

1.4: D In the missing square, one stream enters from the left (at 0.55) and curves down to the bottom (at 0.30), one enters from the top (at 0.45) and curves to the upper-right (at 0.70), and a third connects the bottom-right (0.75) to the lower-right edge (0.25). Only tile D matches all six boundary crossings.

1.5: D In the center square, the two slanting tower walls connect (0.35, 0.65) on the top edge to (0.25, 0.75) on the bottom edge, the cliff meets the left wall at height 0.40, and the two pier lines exit the right edge at 0.20 and 0.55. Only tile D matches all of these edges.